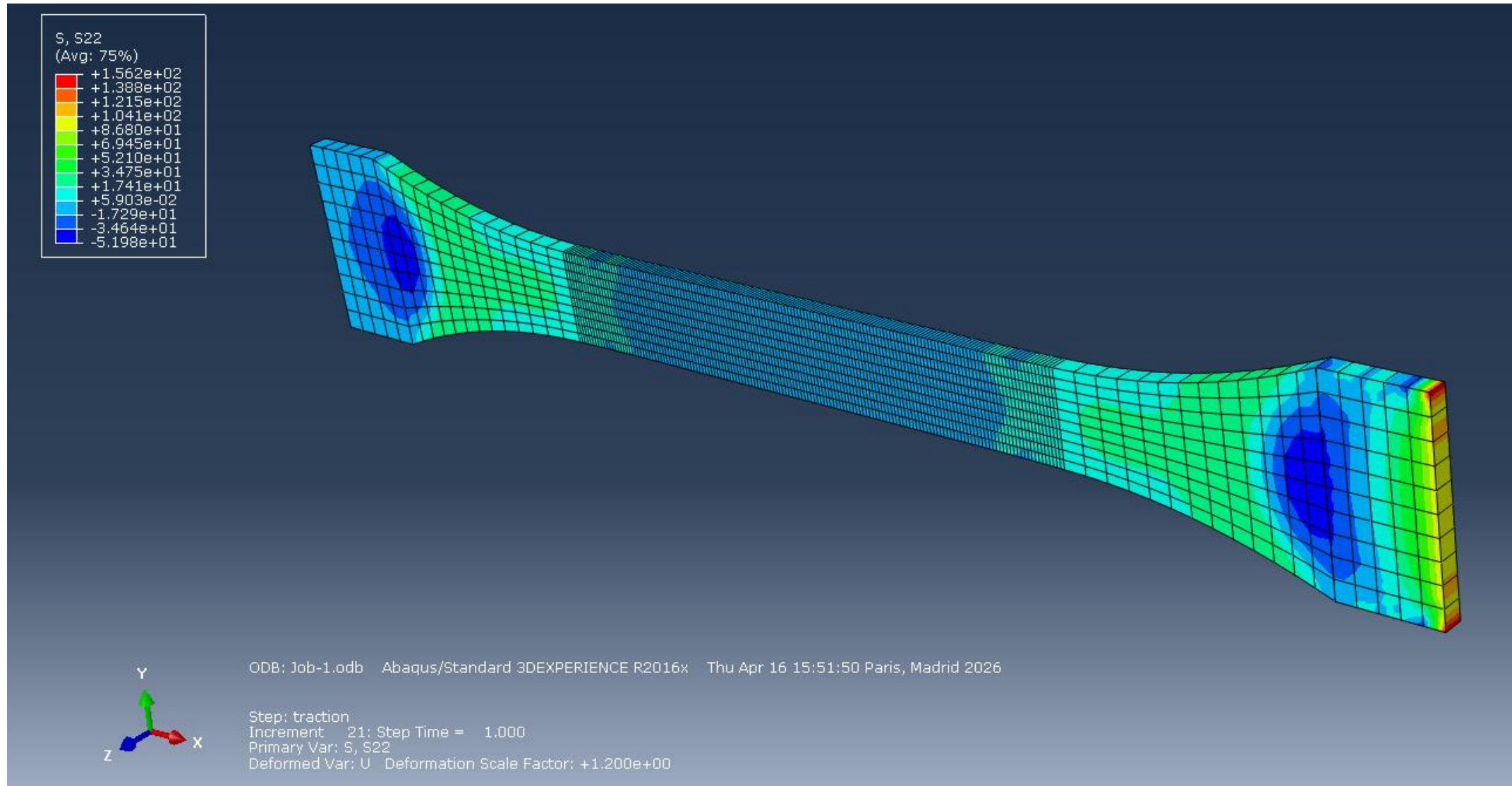
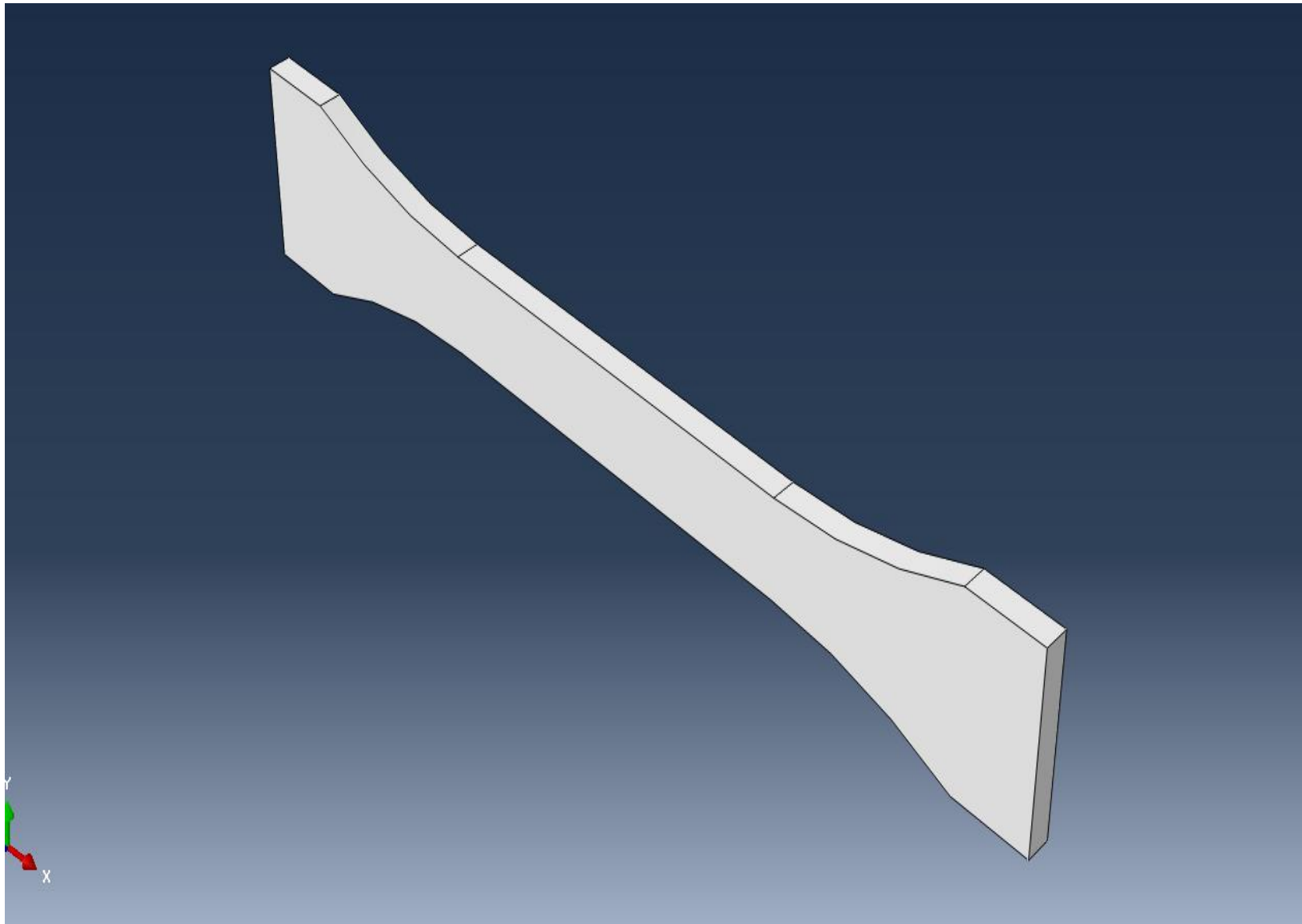


Quand la traction cesse-t-elle d'être uniaxiale ?



De l'éprouvette réelle au modèle numérique 3D



Un matériau élasto-plastique avec écrouissage isotrope

Edit Material

Name: Acier

Description:

Material Behaviors

Elastic

Plastic

General Mechanical Thermal Electrical/Magnetic Other

Plastic

Hardening: Isotropic

☐ Use strain-rate-dependent data

☐ Use temperature-dependent data

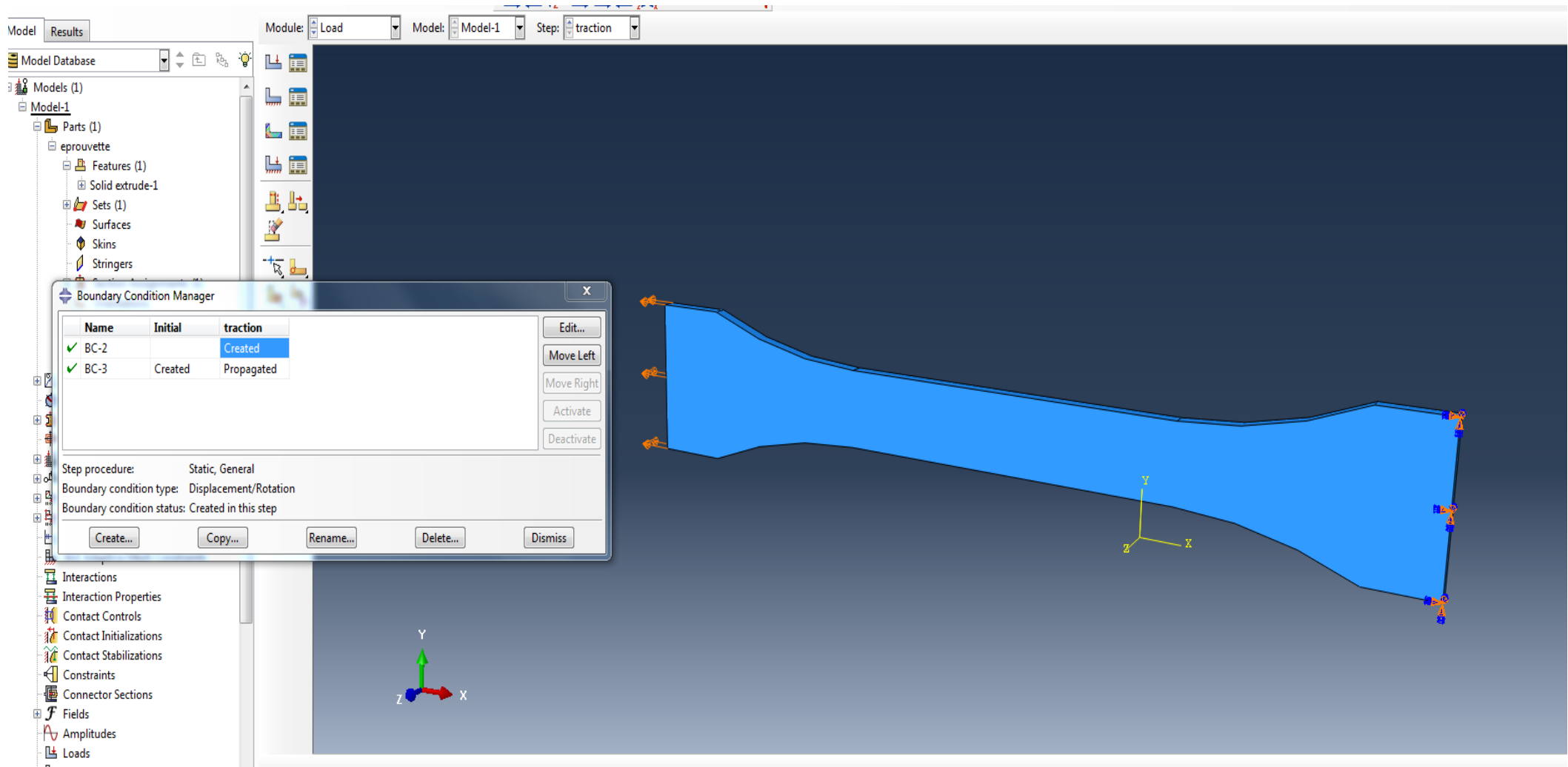
Number of field variables: 0

Data

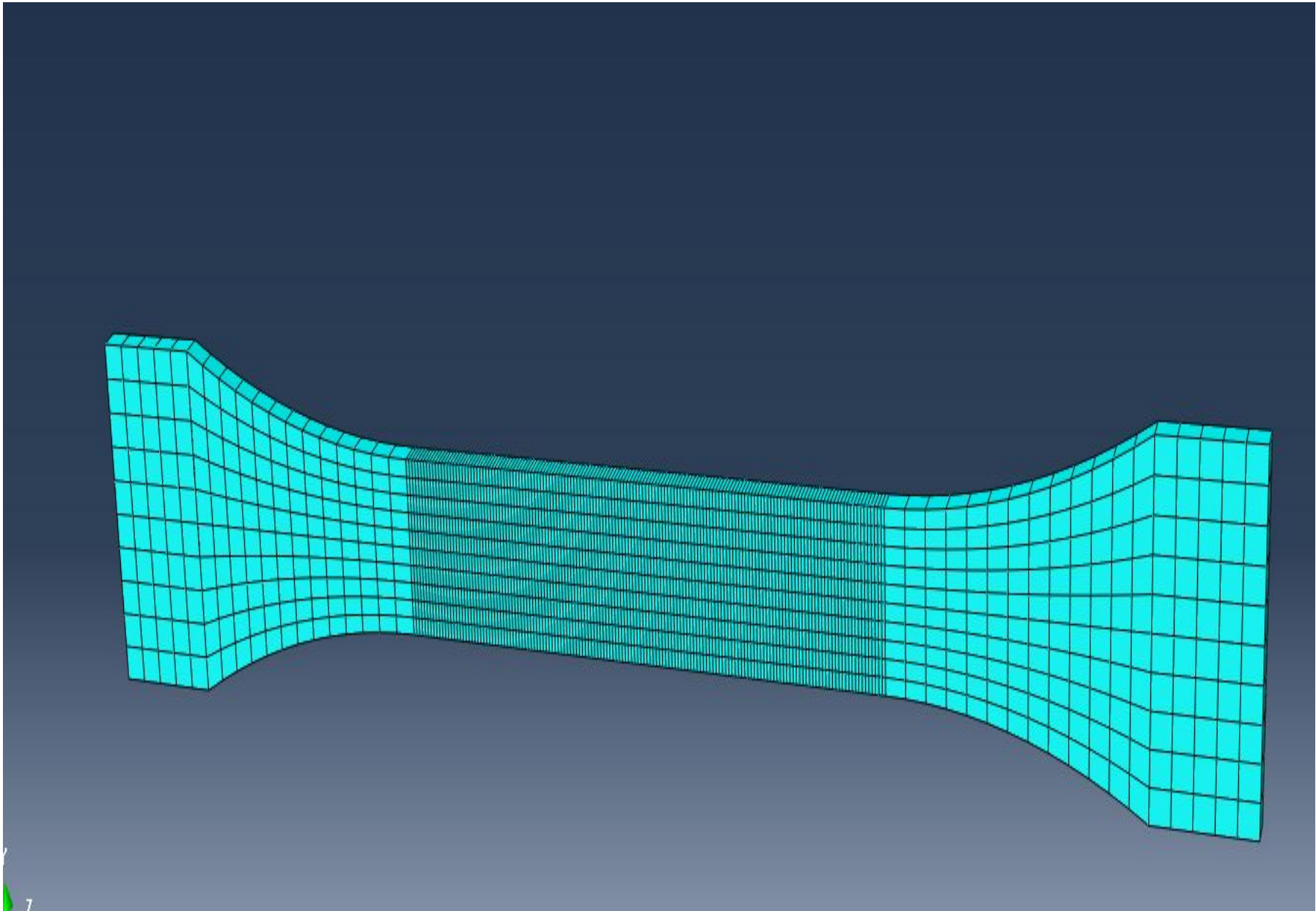
	Yield Stress	Plastic Strain
1	363.4	0
2	393.8	0.0045
3	420.7	0.0121
4	445.5	0.0208
5	480.4	0.0363
6	509.1	0.0528
7	532	0.0695
8	550.4	0.0862
9	565.4	0.1028

OK Cancel

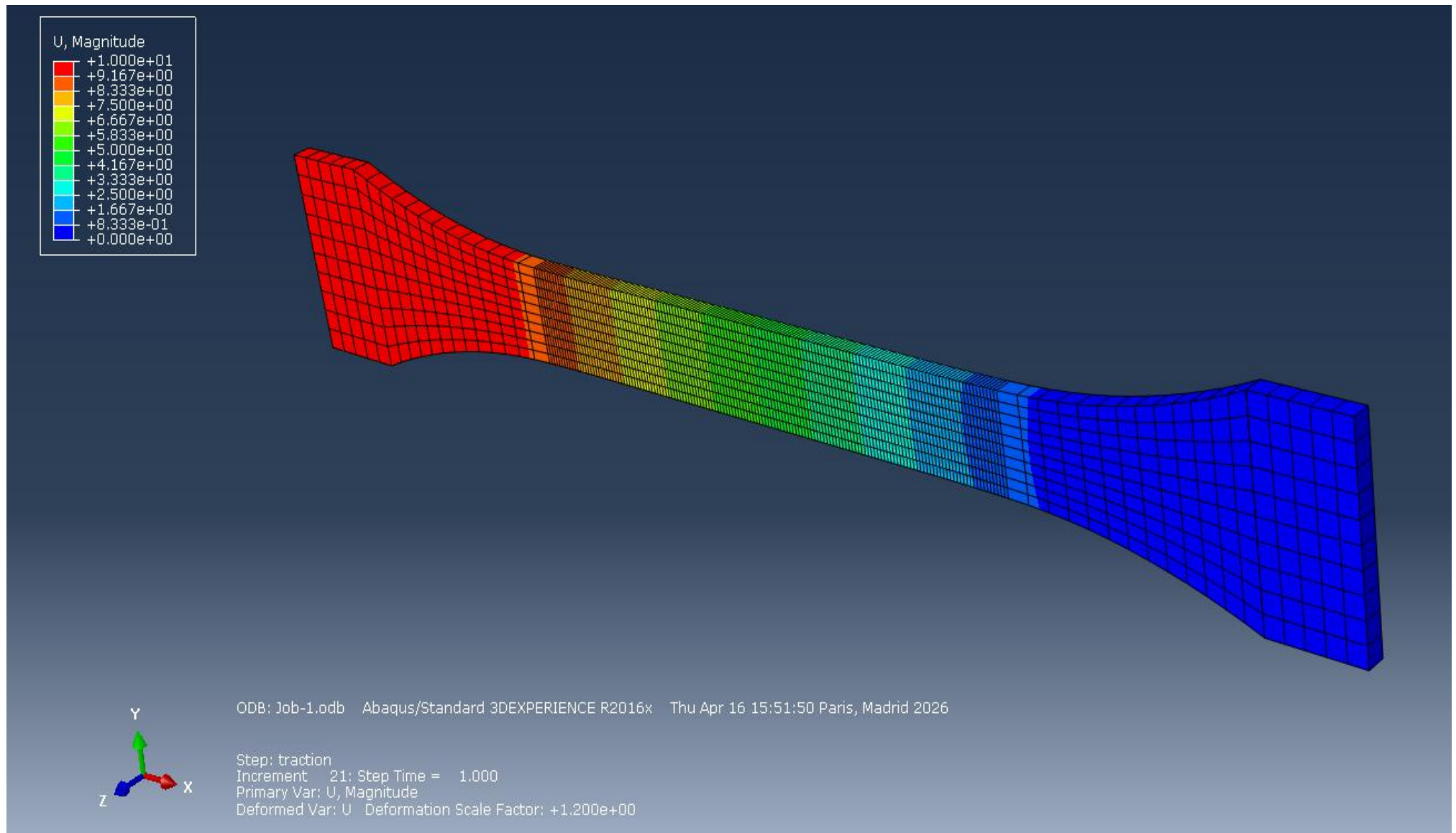
Reproduire l'essai de traction sous Abaqus



Un maillage structuré pour mieux lire les champs

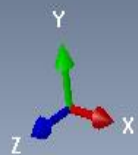
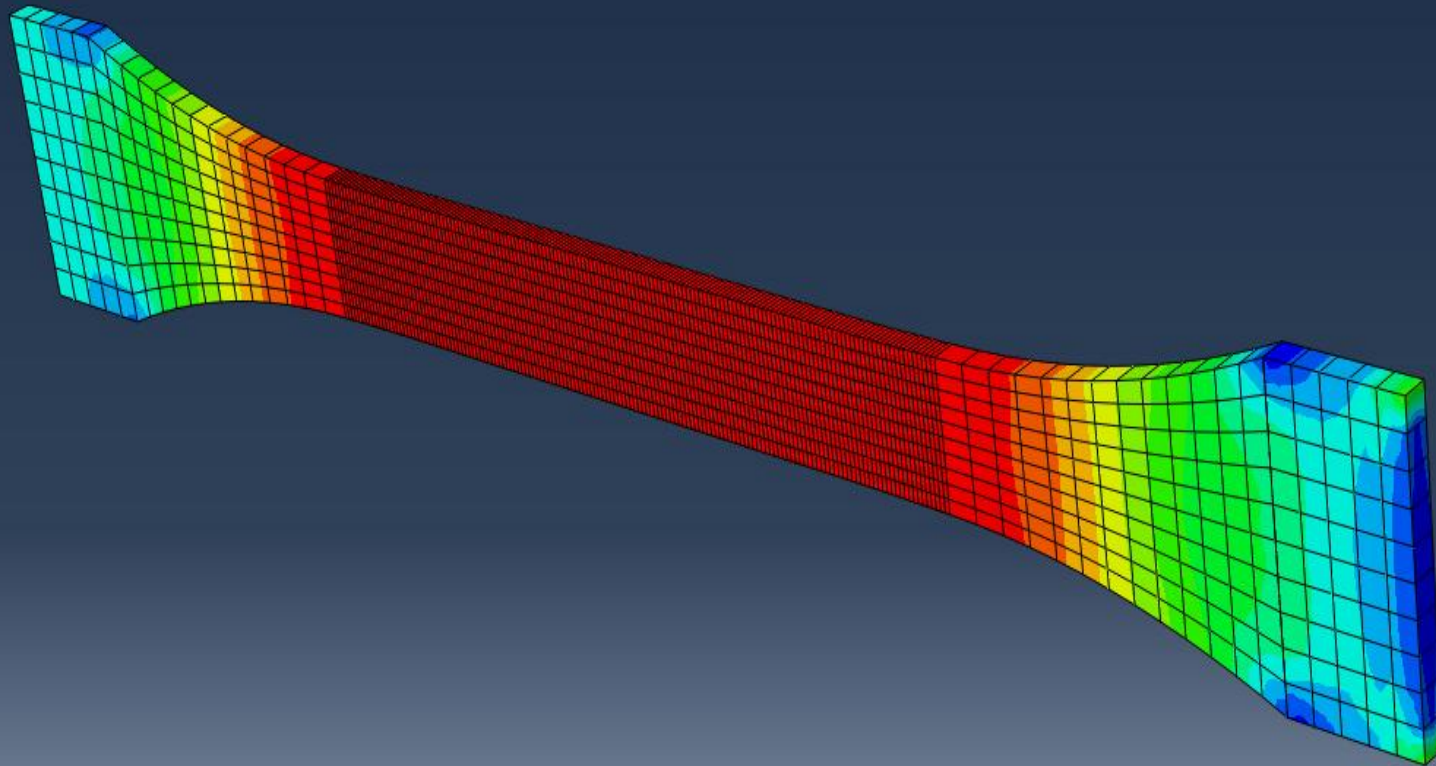
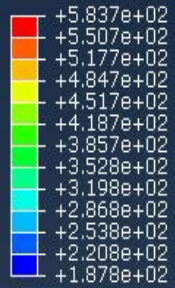


Le déplacement confirme la cohérence du chargement



Von Mises : repérer les zones les plus sollicitées

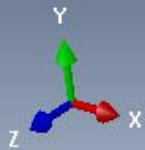
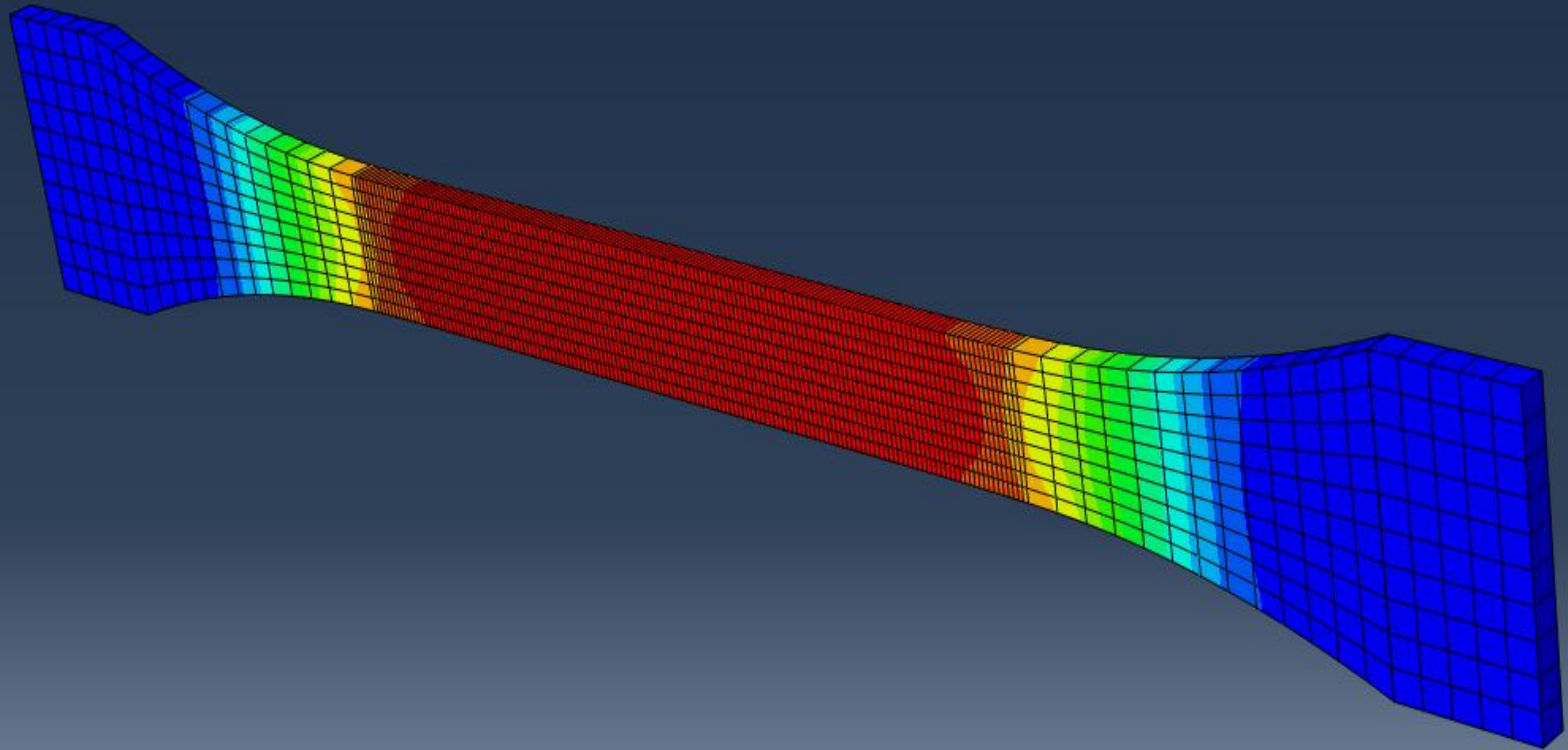
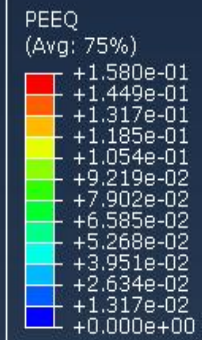
S, Mises
(Avg: 75%)



ODB: Job-1.odb Abaqus/Standard 3DEXPERIENCE R2016x Thu Apr 16 15:51:50 Paris, Madrid 2026

Step: traction
Increment 21: Step Time = 1.000
Primary Var: S, Mises
Deformed Var: U Deformation Scale Factor: +1.200e+00

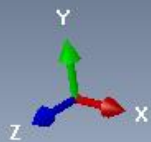
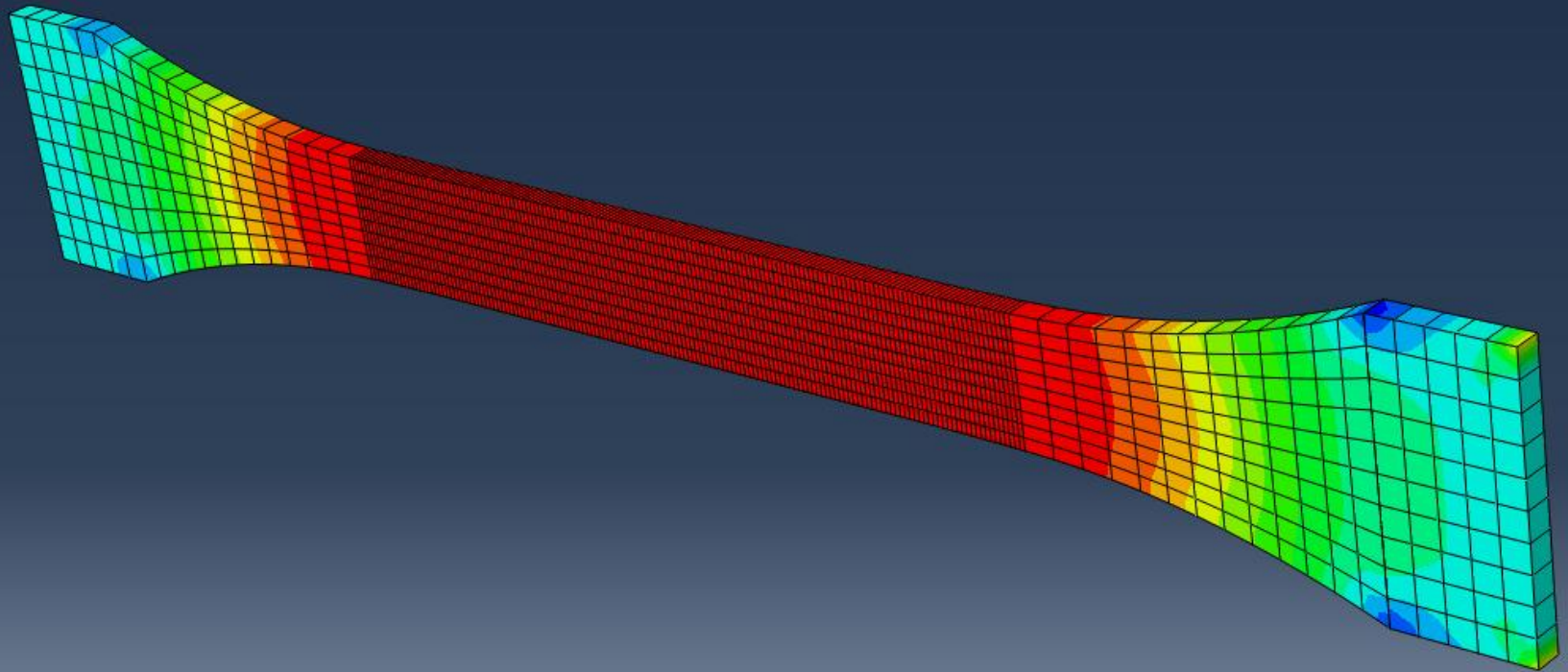
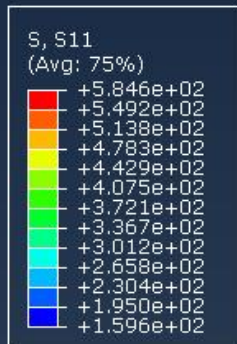
PEEQ : visualiser l'entrée en plasticité



ODB: Job-1.odb Abaqus/Standard 3DEXPERIENCE R2016x Thu Apr 16 15:51:50 Paris, Madrid 2026

Step: traction
Increment 21: Step Time = 1.000
Primary Var: PEEQ
Deformed Var: U Deformation Scale Factor: +1.200e+00

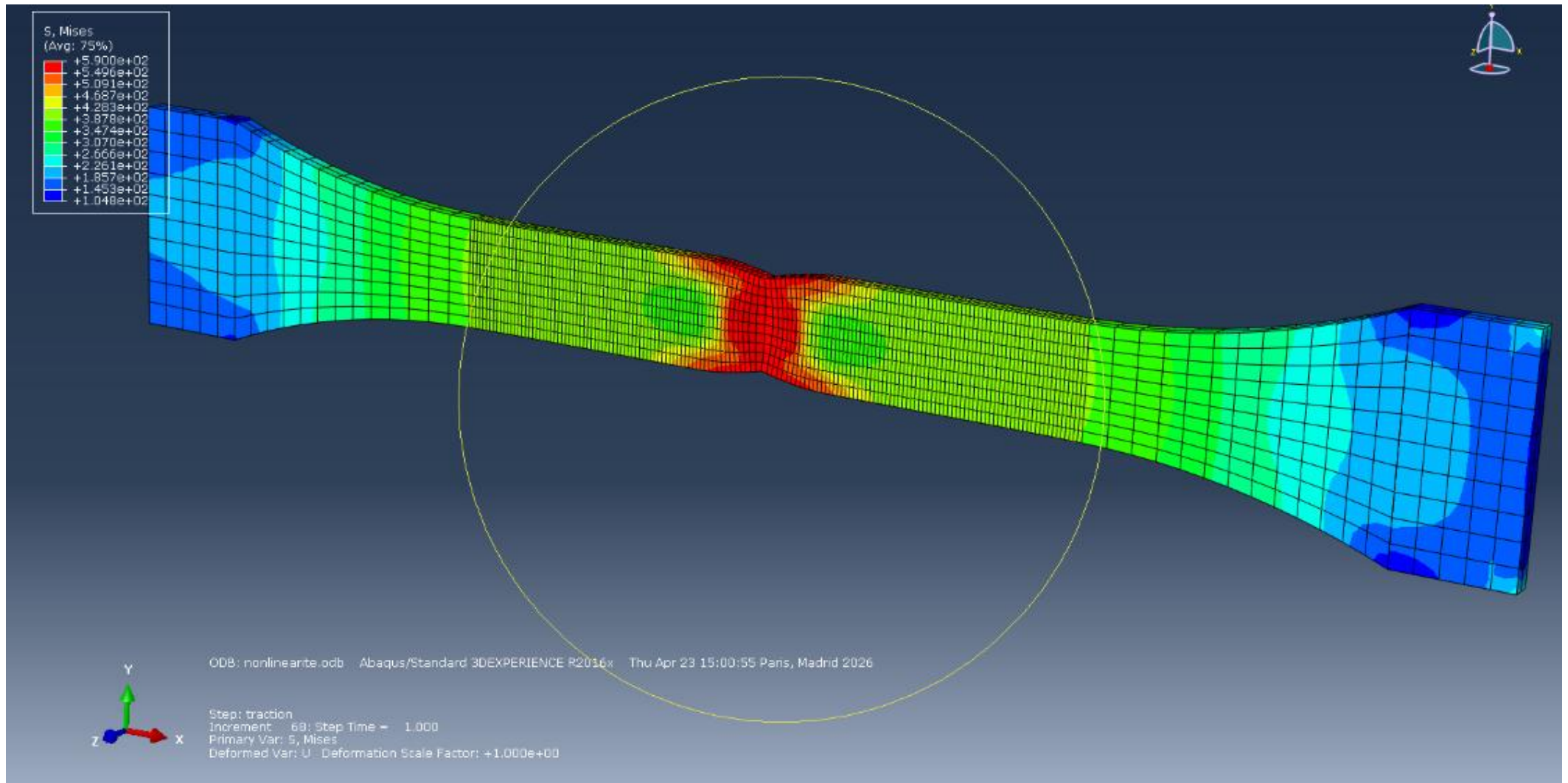
Sans Nlgeom : une plasticité encore diffuse



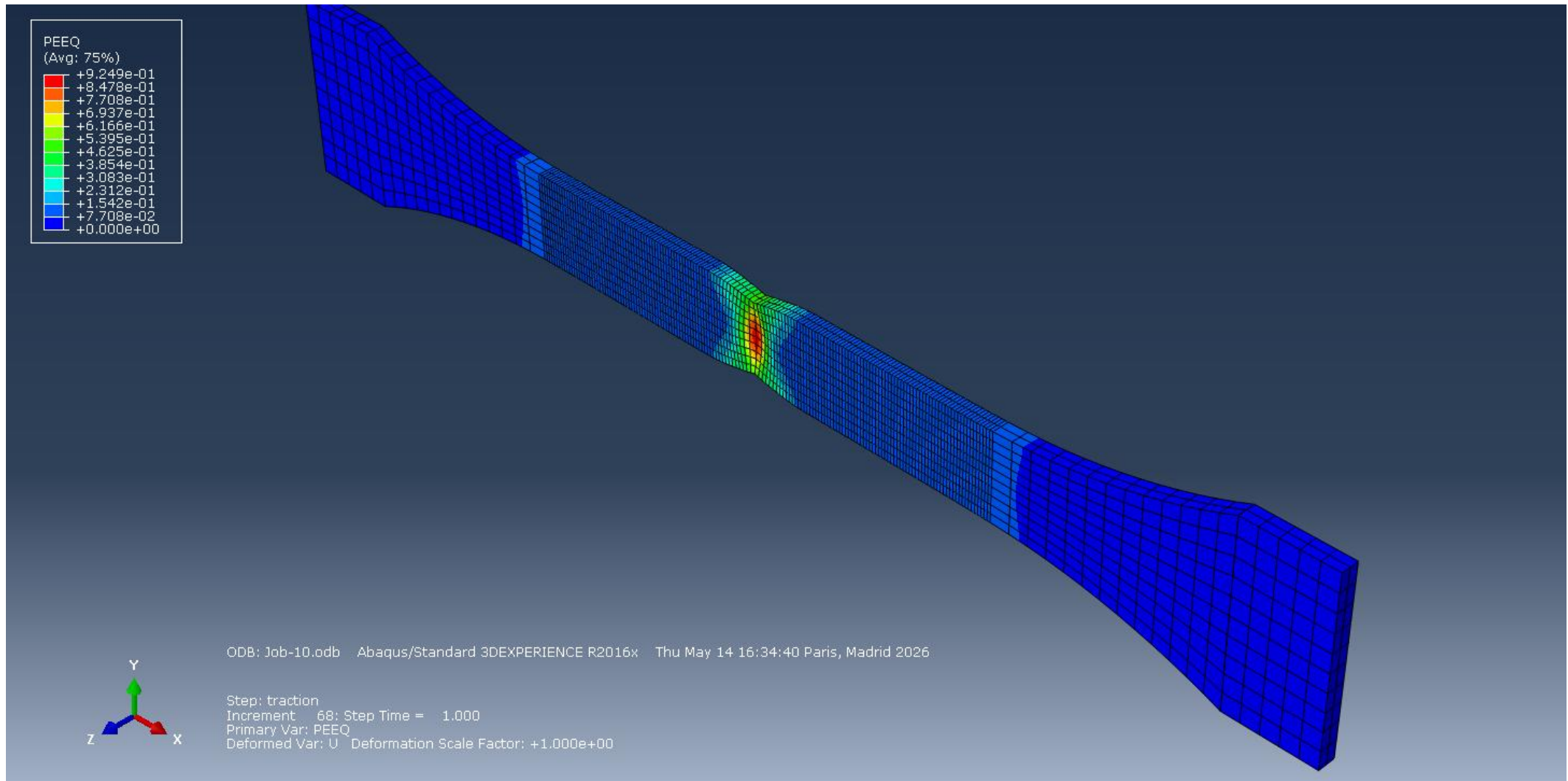
ODB: Job-1.odb Abaqus/Standard 3DEXPERIENCE R2016x Thu Apr 16 15:51:50 Paris, Madrid 2026

Step: traction
Increment 21: Step Time = 1.000
Primary Var: S, S11
Deformed Var: U Deformation Scale Factor: +1.200e+00

Avec Nlgeom : la striction apparaît



Après striction : l'état de contrainte devient localement complexe



Quand la plasticité devient aussi thermique

Courbe Contrainte = f(Déformation)
cas du couplage thermo-mécanique

